



Metamodeling of Alzheimer's Disease

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Abstract

Alzheimer’s Disease (AD) is a highly heterogeneous and multifaceted neurological disorder. By focusing mainly on isolated biological or clinical domains, traditional models usually fail to capture its full complexity. We attempted to overcome these limitations by developing a framework for a unifying, adaptive Bayesian metamodel of the disease. Using a “divide and conquer” approach, we coupled two exemplar models from the literature: a temporal kinetic model and a spatial progression subtyping model. First, we converted the models into fast probabilistic emulators to replicate their output at a fraction of the computational cost. Then, by establishing probabilistic links as constraints between the different models’ variables, we unified them into a joint probabilistic space to force bidirectional information flow. This allowed us to evaluate their joint distribution on simulated patients using a relatively simple MCMC process. The metamodel successfully resolved most conflicting patient data by shifting predictions to satisfy the overall data profile, and amplified predictions when both models aligned. However, not all shifts were as expected, and the joint sampling introduced unexpected posterior dispersion. Therefore, the framework requires further review using empirical patient data. Nevertheless, this work successfully verifies the proof of concept for the metamodeling framework. We created a dynamic, modular metamodel that can also serve as a powerful cross-domain evaluator and can easily be expanded using empirical data to facilitate the discovery of multi-dimensional AD phenotypes.

Introduction

Alzheimer's Disease

One of the most infamous neurodegenerative diseases, which is also the most common cause of dementia[1], is AD. Its definition keeps changing over the years, yet it is increasingly recognized as a heterogeneous, multifactorial and multifaceted disorder, associated with neurological, psychiatric, and systemic comorbidities. It operates across several interconnected domains, including cognitive-behavioral symptomatology, underlying biological mechanisms, varying neuroanatomical networks, and complex genetic variability. These vast complexity and heterogeneity are the root cause of the ongoing struggle to understand the disease and develop suitable treatments for different patients.

Historically, research and clinical diagnosis of AD have focused heavily on core hall-mark biomarkers, like the abnormal extracellular accumulation of Amyloid-Beta ($A\beta$) protein plaques and the intracellular aggregation of hyperphosphorylated Tau protein tangles[2]. While $A\beta$ often marks early pathological changes, the spatial spread of Tau pathology correlates more directly with localized neurodegeneration and cognitive decline. Adding to this complexity is genetic variability and genetic profiles known to influence the disease, such as carrying the Apolipoprotein E4 (APOE4) allele, which stands out as the strongest genetic risk factor, significantly accelerating pathology onset and altering disease progression across different patient populations[3].

Traditional diagnostic and predictive frameworks usually rely on limited-field mechanisms, such as the biological markers we mentioned, cellular characteristics and inter-cellular interactions, behavioral or anatomical characteristics, or genetic factors. Understanding the interplay between these diverse changes in AD is essential to define and identify disease phenotypes for targeted interventions, all while available disease data tends to be noisy, sparse and spread across entirely different scales and measures. Therefore, a suitable unifying methodology/framework is required to handle said diversity. Such a framework should be capable of connecting the different faces of

the disease and generating reliable and extensive predictions of how diverse factors drive disease progression.

The Metamodeling Approach

This presents a computational challenge. Forcing every dimension of the AD pathology into a single mathematical representation is inherently impractical. We suggest and examine here a divide-and-conquer strategy: **Bayesian metamodeling**. It is based on utilizing smaller, specialized models that each capture a particular facet of AD, and then fuse/couple them together under a Bayesian probabilistic framework - the metamodel[4]. This strategy allows a highly adaptive model. The metamodel can continuously evolve alongside new clinical and biological discoveries, on two fronts: first, by having the flexibility to change, update or replace the individual submodels themselves; and second, by dynamically adjusting the probabilistic connections between the submodels, which are based on current scientific knowledge[5]. Thus, discoveries from one specific field can influence the mathematical constraints of other domains, ensuring that our computational tools can continuously adapt to the increasingly complex knowledge of the disease - which makes this strategy highly suitable to the multifaceted and heterogeneous problems of AD.

Exploring the Bayesian metamodeling strategy offers computational and modeling advantages for analyzing any complex system, not limited to AD, diseases or even biology in general. It overcomes the rigidity of monolithic architectures, by allowing different computational approaches, such as mechanistic equations and data-driven algorithms to interact and enrich each other under a unifying framework. Furthermore, understanding the use of probabilistic tools provides a beneficial way to manage uncertainty and reduce expensive model computations.

Here, we attempted to create a metamodel of characterization and prediction of AD progression in patients that combines two partial models, corresponding to two different essential aspects of understanding the disease for specific patients. The first partial model describes the temporal dynamics, kinetic rates and clinical decline rates

of biomarkers[6]; and the second identifies distinct spatiotemporal trajectories of cortical tau deposition to stratify patients into phenotypic subtypes and progression trajectories[7]. Both models are probabilistic, and we aim to show that, when formulated within a probabilistic framework, the unified metamodel combines their distinct representations to calibrate prediction certainties and inferences, yielding more robust and reliable forecasts of AD progression.

Results

We explored the Bayesian metamodeling approach for Alzheimer’s Disease progression using two known models from the literature. In order to utilize them in the Bayesian metamodeling framework, we converted each model to a probabilistic surrogate model, so that we could couple them and perform inference and sampling in a joint probabilistic space.

Thus, our Bayesian metamodeling pipeline comprises three main stages:

1. Individual model surrogate training.
2. Creating a coupling architecture.
3. Sampling and inference from the joint coupled probabilistic framework.

1. Individual Model Surrogate Training

General

Each surrogate model we used is a fast, probabilistic emulator that replaces an expensive computational model with a probability density function. Mathematically, it models the conditional probability distribution of the model outputs given the inputs:

$$p(\Theta|\mathbf{x})$$

where

- $\mathbf{x}$ - inputs: parameters, initial conditions, patient baseline characteristics and measurements.
- Θ - outputs: predicted future states, disease progression coefficients, biomarker levels, classification probabilities, etc.

Thus the trained surrogate evaluates the probability density of a given prediction Θ and can also draw samples from the conditional output distribution of the model $\Theta \sim p(\Theta|\mathbf{x})$.

The surrogate model could be obtained by any approach for modeling statistical distributions, where the major tradeoffs between methods are often expressiveness and interpretability[4]. Since we are dealing with complex models, we chose to focus on the expressiveness of the surrogates and implemented both surrogates using modern simulation-based inference (SBI[8]) methods. Specifically, using the original model we created a large set of pairs of inputs $\mathbf{x}$ and their corresponding outputs Θ , which formed the training dataset. Then, we used a neural posterior estimator (NPE) to fit a deep generative neural network directly to the dataset. This network approximates the true distribution of the model $p(\Theta|\mathbf{x})$.

Full simulation outputs might be high-dimensional and complex, and their internal mechanics might make full-state probability modeling computationally hard or intractable. Therefore, selecting specific inputs and outputs for the surrogate to model can serve as an essential solution that defines a compact and usable probabilistic description. The surrogates only need to learn the essential causal relationships of the original complex model’s inputs-output relationships relevant to the broader meta-model. The selection of variables for the surrogate is critical, as it establishes what information propagates forward and backward within the metamodel, and should be done with the later coupling and forward and backward propagation of information between models in mind.

The Models

First Model - The ODE model The first and main model is the ODE-based model of AD presented by Bossa and Sahli in “A multidimensional ODE-based model of Alzheimer’s disease progression”[6]. It uses longitudinal differential equations to track a state vector of continuous disease-burden variables, like Amyloid-Beta, Tau, and cognitive domain scores. Using personalized Bayesian fitting, the model infers

probability distributions over the differential equation rate matrix from the patient's data, estimating each parameter as a distribution rather than a single fixed value. It then uses these parameters to forecast future trajectories of the variables. Importantly, it also uses an ordered logistic GLM to calculate a weighted disease burden score, which is used to assign probabilities for clinical diagnosis of AD severity: Cognitive Normal (CN), Mild Cognitive Impairment (MCI), and Dementia (AD).

Simply put, its inputs are a patient's demographics and their observed longitudinal biomarker and cognitive measurements; and its outputs are the estimated rate distributions, as well as trajectory forecasts for future biomarkers levels and clinical diagnostic probabilities.

To replace it with a probabilistic surrogate model, we chose to focus only on some of the parameters:

- **Surrogate inputs:** Age, APOE4 status, Amyloid-Beta baseline + 2-year measure, Tau baseline + 2-year measure.
- **Surrogate outputs:** Tau self-dynamics rate, Amyloid-Beta self-dynamics rate, Amyloid-Beta effect on Tau rate, cognitive memory result baseline + 5 year, clinical stage result baseline + 5 year.

We successfully trained a surrogate model over data derived from the original model, as can be observed in the figures below that compare both the data distributions in general, and specific patients' predictions.

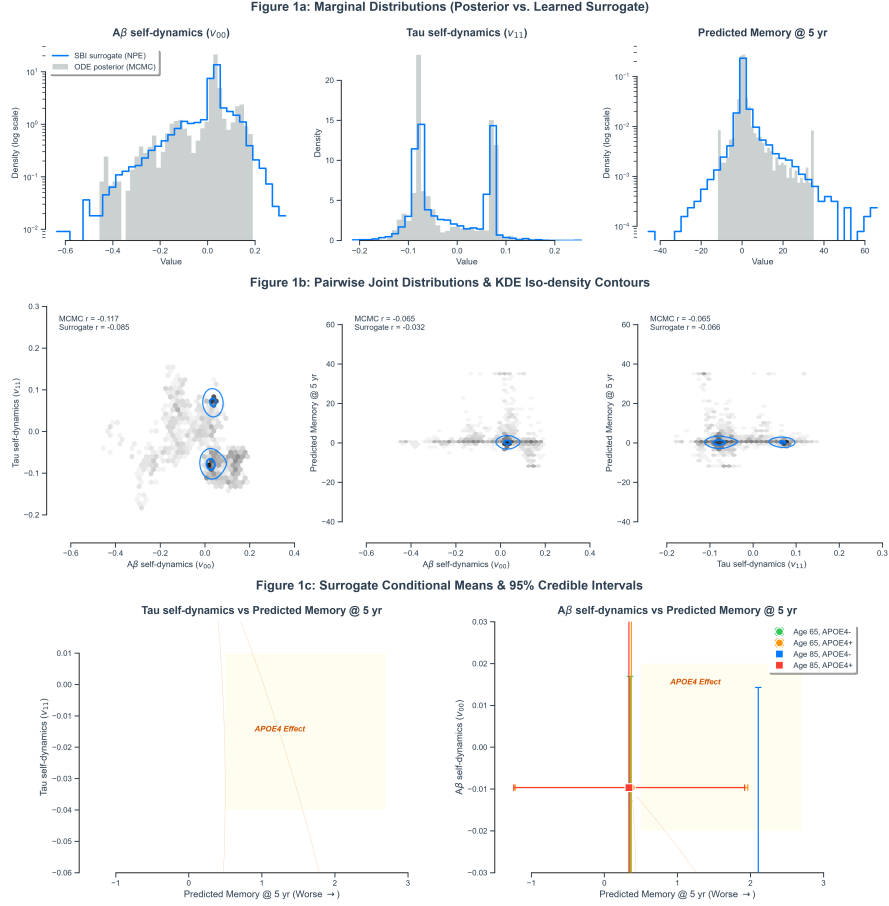


Figure 1: **Validation and performance of the surrogate model Neural Posterior Estimator (NPE) surrogate against original ODE MCMC baseline samples.** (A) **Marginal Distributions (Posterior vs. Learned Surrogate).** One-dimensional marginal probability densities comparing ground-truth MCMC posterior samples from the continuous-time ODE model (gray shaded histograms) against sample draws generated by the Masked Autoregressive Flow (MAF) NPE surrogate (blue stepped lines). Plots depict A β self-dynamics (v_{00} , log-scale density), Tau self-dynamics (v_{11} , linear density), and 5-year forecasted Memory deficit scores (log-scale density).

(B) **Pairwise Joint Distributions and Density Contours.** Two-dimensional joint distributions of MCMC samples (gray points/hexbins) overlaid with surrogate kernel density estimation (KDE) contours enclosing the 50% and 90% highest density intervals (blue lines). Pearson correlation coefficients (r) for MCMC baselines and surrogate draws are displayed in the upper-left of each panel.

(C) **Surrogate Conditional Means and 95% Credible Intervals.** Conditional inference generated by the surrogate for four distinct patient archetypes: Age 65 / APOE4 $-$ (green circle), Age 65 / APOE4 $+$ (orange circle), Age 85 / APOE4 $-$ (blue square), and Age 85 / APOE4 $+$ (red square). Point estimates represent conditional means, with horizontal and vertical error bars indicating 95% credible intervals (CIs). The soft yellow highlighted area (APOE4 Effect) denotes the systematic pathological shift in kinetic rate constants and cognitive decline driven by APOE4 carrier status.

Individual Validation: Biological Self-Dynamics
ODE Posterior MCMC vs. Learned NPE Surrogate

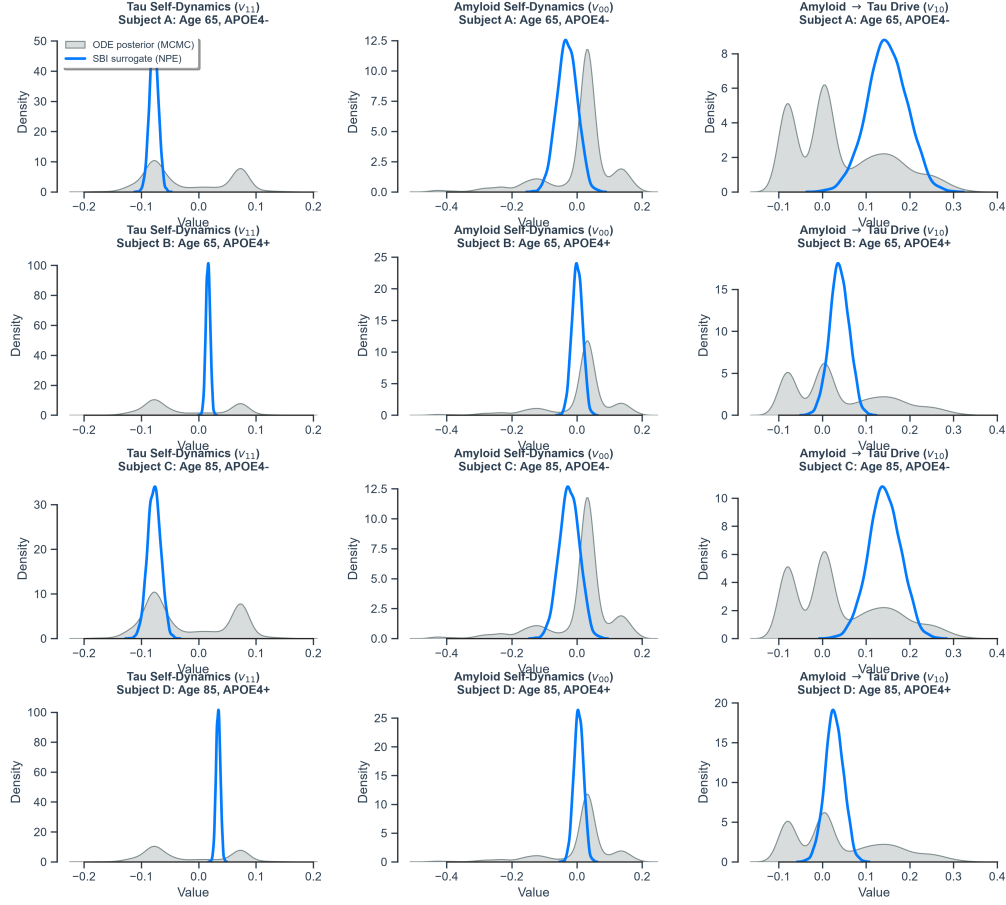


Figure 2: **Individual Subject Validation: Biological Kinetic Parameters.** Comparison of ground-truth continuous-time ODE MCMC posterior distributions (gray shaded densities) against sample draws generated by the SBI NPE surrogate (solid blue lines) across four distinct patient archetypes:

Subject A: Age 65, APOE4- (Younger, low genetic risk)

Subject B: Age 65, APOE4+ (Younger, high genetic risk)

Subject C: Age 85, APOE4- (Older, low genetic risk)

Subject D: Age 85, APOE4+ (Older, high genetic risk) Columns depict the posterior probability densities for three key differential equation parameters: **(Left)** Tau self-dynamics velocity (v_{11}), **(Center)** $A\beta$ self-dynamics velocity (v_{00}), and **(Right)** $A\beta \rightarrow$ Tau causal driving rate (v_{10}).

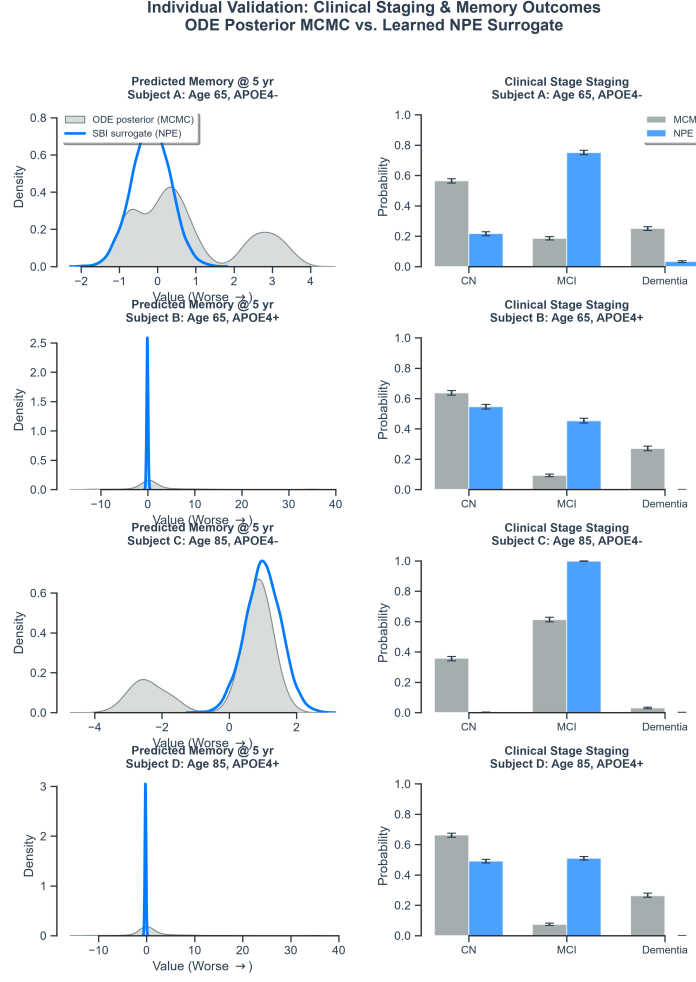


Figure 3: **Individual Subject Validation: 5-Year Cognitive Forecasts and Diagnostic Staging.** Validation of 5-year longitudinal clinical predictions for Subjects A–D. **(Left Column)** Density distributions of 5-year predicted memory deficit scores (where higher values indicate worse cognitive impairment), comparing MCMC baselines (gray shaded) against surrogate predictions (blue line). **(Right Column)** Probability bar charts for diagnostic category conversion at 5 years: **CN** (Cognitively Normal), **MCI** (Mild Cognitive Impairment), and **AD** (Dementia). Gray bars represent ground-truth MCMC GLM classifications; blue bars represent NPE surrogate predictions. Whiskers indicate 95% bootstrap confidence intervals.

Second Model - The SuStaIn model The second model we used is the spatial progression model presented by Vogel et al. in "Four distinct trajectories of tau deposition identified in Alzheimer’s disease"[7]. It is a subtype and stage inference (SuStaIn[9]) model that focuses on the Tau biomarker protein’s pathology. It maps cross-sectional biomarker accumulation across multiple brain regions into

distinct temporal pathways (subtypes) and sequential disease severity stages. Using Bayesian MCMC, the model automatically discovers the distinct trajectory sequences and calculates a patient’s probabilistic placement across these subtypes and stages. Crucially, rather than assuming a single uniform progression path, it estimates the posterior probability of a patient belonging to each subtype and occupying each pathological stage. That is, it first uses the entire training data to infer and characterize the different possible subtypes of Tau progression/trajectory in AD, and then it can be used to assign patients to a subtype and their stage in it.

Simply put, its inputs are a patient’s cross-sectional regional Tau measurements across different brain regions, and its outputs are the posterior probability distributions over disease subtypes and discrete progression stages.

To replace it with a simpler probabilistic surrogate model, we chose to focus on the regional Tau scores and collapsed the discrete stage probabilities into a continuous expected value that represents the stage within the trajectory:

- **Surrogate inputs:** Regional Tau PET Z-scores across brain regions.
- **Surrogate outputs:** A probability of belonging to each of the subtypes, and a single continuous expected spatial stage.

Here as well, we successfully trained the neural posterior estimator as a surrogate model. We used simulated data with 7 stages for each subtype and 3 Tau trajectory subtypes overall. We can observe its results and performance in the graphs below.

Figure 1: Validation and Performance of the pySuStain Flow-Based NPE Surrogate
Panel A: Marginal Densities (Fidelity Evaluation)

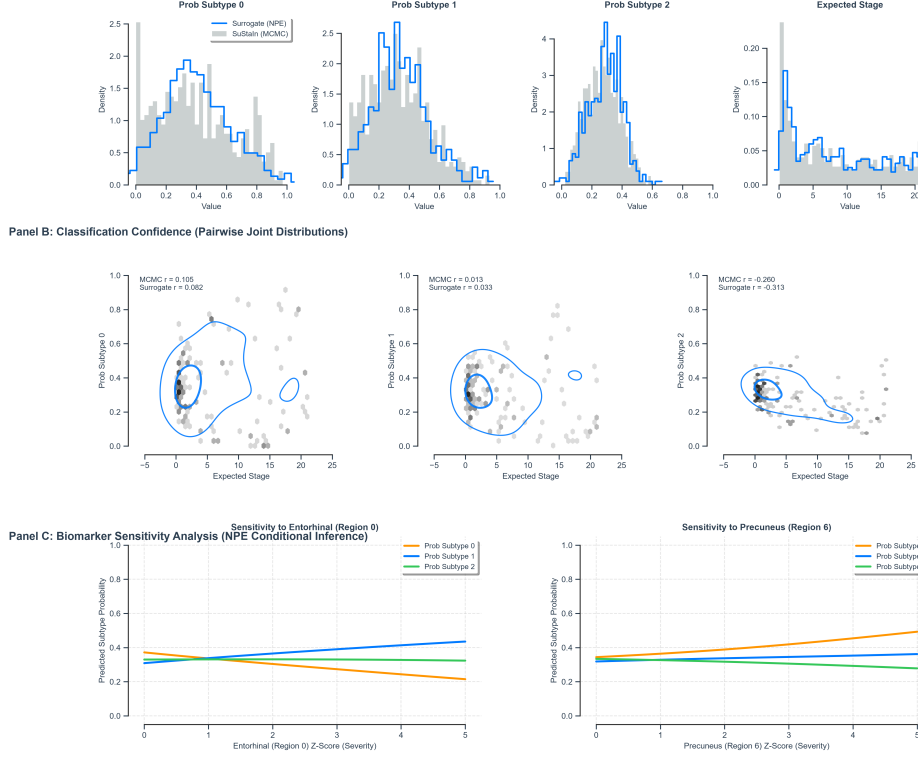


Figure 4: **Performance and validation of the SuStaIn Neural Posterior Estimator (NPE) surrogate against original model baselines.**

(A) Marginal Densities (Fidelity Evaluation). One-dimensional marginal probability density distributions comparing ground-truth MCMC sample outputs from pySuStaIn (gray shaded histograms) against sample draws generated by the NPE surrogate (blue stepped lines). Panels depict Subtype 0 probability, Subtype 1 probability, Subtype 2 probability, and continuous expected spatial stage (0–21).

(B) Classification Confidence and Pairwise Joint Distributions. Two-dimensional joint probability distributions plotting continuous expected spatial stage (x-axis) against individual subtype classification probabilities (y-axis). Ground-truth MCMC samples are shown as log-binned hexbins (gray), overlaid with surrogate kernel density estimation (KDE) contours enclosing the 50% and 90% highest-density intervals (blue lines). Displayed Pearson correlation coefficients (r) confirm accurate preservation of joint covariance.

(C) Biomarker Sensitivity Analysis (NPE Conditional Inference). Conditional subtype classification trajectories under regional severity sweeps. Z-scores for Region 0 (Entorhinal cortex, classic limbic origin) and Region 6 (Precuneus, atypical neocortical origin) are independently swept from 0.0 to 5.0 while holding non-target regions at baseline (0.5 0.5). Curves trace the dynamic transition probabilities for Subtype 0 (orange), Subtype 1 (blue), and Subtype 2 (green).

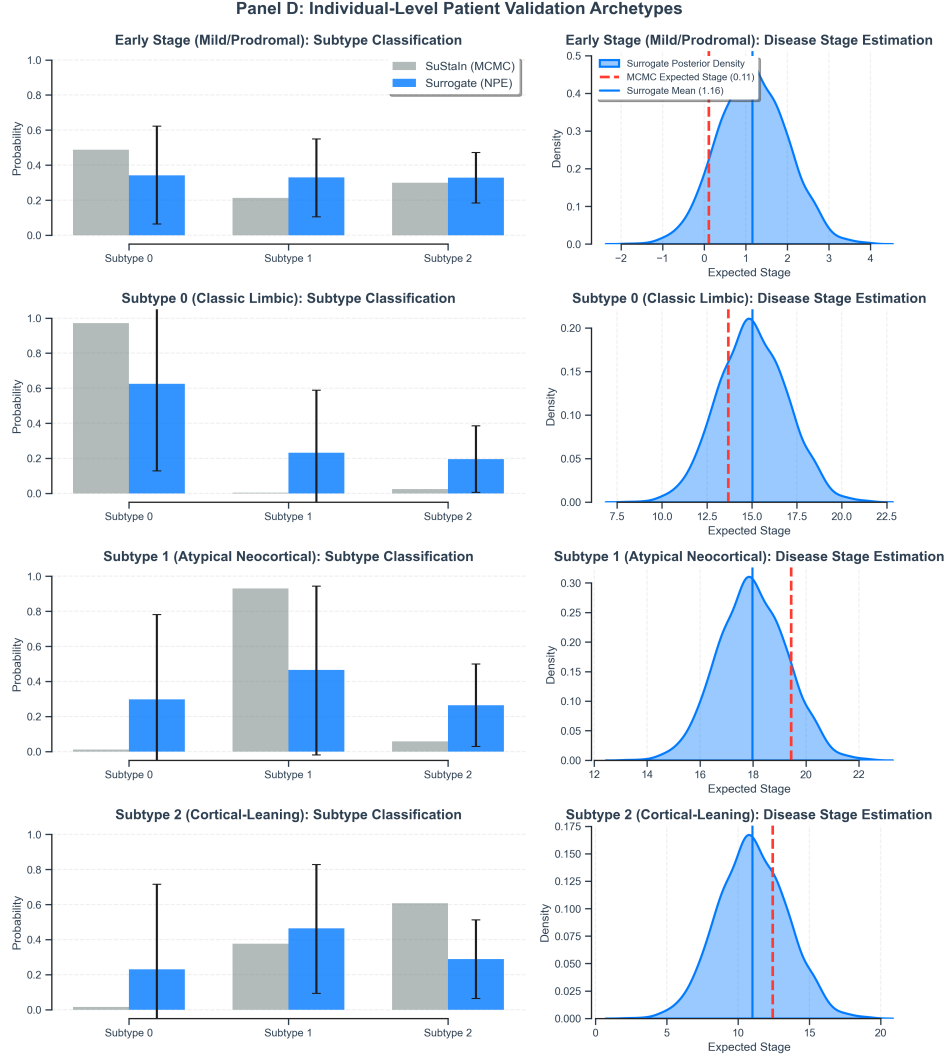


Figure 5: Individual Subject Validation across Representative Patient Archetypes. Validation of the SuStaIn NPE surrogate against ground-truth original model (MCMC) baseline outputs for three synthetic patient archetypes representing distinct spatial progression profiles:

Row 1: Early Stage (Mild/Prodromal) - Patient at early disease onset with low overall pathology.

Row 2: Subtype 0 (Classic Limbic) - Patient exhibiting strong medial temporal/limbic Z-score signatures.

Row 3: Subtype 1 (Atypical Neocortical) - Patient exhibiting strong neocortical Z-score signatures.

Row 4: Subtype 2 (Cortical-Leaning) - Patient exhibiting cortical-leaning Z-score signatures.

(Left Column) Subtype Classification Probabilities. Comparison of ground-truth pySuStaIn MCMC subtype probabilities (gray bars) against sample means predicted by the NPE surrogate (blue bars). Error bars on surrogate estimates represent 95% Bayesian credible intervals (CIs).

(Right Column) Disease Stage Estimation. Continuous posterior probability density curves of the surrogate's estimated spatial stage (expected_stage, blue shaded KDE curves). Vertical dashed red lines indicate ground-truth MCMC expected stages; vertical solid blue lines indicate surrogate posterior mean predictions.

2. Creating a Coupling Architecture

The General Coupling Framework

In order to fuse the models and couple them within one metamodel framework, we had to make them operate in the same probability space: instead of calculating the probabilities of the outputs of the ODE model ($p(\Theta_{ODE}|\mathbf{X}_{ODE})$) and the SuStaIn model ($p(\Theta_{SuStaIn}|\mathbf{X}_{SuStaIn})$) separately and independently, we needed to calculate their joint probability: $p(\Theta_{ODE}, \Theta_{SuStaIn}|\mathbf{X}_{shared})$.

To do so, we created links between different variables of the two surrogate models. Those links push the models' variables to agree - whether both models share an identical variable whose values must converge; or variables from both models share a mathematical or empirical relationship, and their values should sustain a certain relation or link. These coupling links (C) represent domain constraints imposed on the surrogates, derived from a meta view using real-world knowledge and biological and medical premises or conclusions across external studies. Thus, they act as constraints on the joint distribution of the surrogates, and links' probabilities depend on their outputs. That is, the greater the alignment between the two sub-models, the greater the probability that the coupling link constraint is satisfied.

This allows us, after applying the coupling constraints, to recalculate the joint probability distribution function, using Bayes rule:

Applying the constraints, $p(\Theta_{ODE}, \Theta_{SuStaIn})$ becomes $p(\Theta_{ODE}, \Theta_{SuStaIn}|C)$. Using Bayes and our assumption that both surrogates are independent at first (two different models (we omitted the obvious conditions of the inputs $\mathbf{X}$ for comfort):

$$\begin{aligned} p(\Theta_{ODE}, \Theta_{SuStaIn}) &= \frac{p(C|\Theta_{ODE}, \Theta_{SuStaIn}) \cdot p(\Theta_{ODE}, \Theta_{SuStaIn})}{p(C)} \\ &\propto p(C|\Theta_{ODE}, \Theta_{SuStaIn}) \cdot p(\Theta_{ODE}) \cdot p(\Theta_{SuStaIn}) \end{aligned}$$

Since we can calculate both $p(\Theta_{ODE})$, $p(\Theta_{SuStaIn})$ with the surrogate models, it remains to define the probability score $p(C|\Theta_{ODE}, \Theta_{SuStaIn}) := \phi(\Theta_{ODE}, \Theta_{SuStaIn})$.

This can be done in various ways given to our choice, and this is where we defined the coupling itself.

For example, if we wanted to make two specific different variables $\theta_{ODE}, \theta_{SuStaIn}$ to agree on their values, we could define a gaussian link between them, such that

$$\phi(\Theta_{ODE}, \Theta_{SuStaIn}) = \mathcal{N}(\theta_{ODE}; \theta_{SuStaIn}, \sigma) = \mathcal{N}(\theta_{SuStaIn}; \theta_{ODE}, \sigma)$$

where σ determines the strength of the coupling. Clearly the probability increases where they are similar and decreases where the variables are far part; and the bigger σ the greater is the decrease in probability score for each difference size.

Notice that we can concatenate different couplings by treating them as indepedent, so that

$$\phi(\Theta_{ODE}, \Theta_{SuStaIn}) = \phi_1(\Theta_{ODE}, \Theta_{SuStaIn}) \cdot \dots \cdot \phi_n(\Theta_{ODE}, \Theta_{SuStaIn})$$

After that, for any evaluation or sampling method we have over the entire data we want to infer about, we can evaluate it using no less than at least three scores:

- The probability scores of the output as is calculated by each surrogate model separately.
- The probability score given by each coupling constraint we defined.

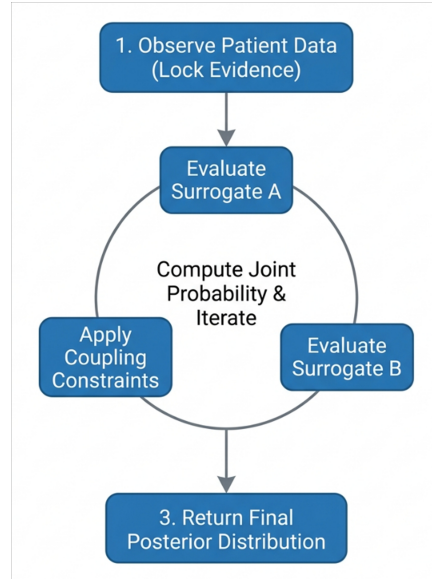


Figure 6: A simple diagram of the information flow in the metamodel.

The Coupling Links

We used three coupling links between the two models, two Gaussian links that operate on transformations of different variables from both surrogate models, and one other kind of link between different variables (a 'directional potential' link). The linked variables under every coupling link can be viewed in Figure 7 at the end of this subsection.

Directional Potential Link - Subtype-Driven Velocity/ τ Dynamics

- **Coupled variables from ODE model:** τ self rate (entry v_{11} in the ODE rate matrix)
- **Coupled Variables from SuStain model:** Subtypes probabilities (probability of subtype 0, probability of subtype 1, probability of subtype 2)

Different real-world subtypes of τ trajectories correspond to distinct τ self spread rates[7], so we inferred **there should be a link between the probabilities of the subtype the SuStaIn model outputs, and the τ rate parameter in the**

ODE model’s output of the rate matrix. While Vogel et al. identified four subtypes, our simulated data included three. Based on the main shared characteristic of our subtypes and theirs, we decided to which subtypes we want to relate with a general faster rate (subtypes 0, 2) and to what degree, and which with a slower Tau spread rate (subtype 1).

Since the values of the spread rate and the probabilities of subtypes operate on completely different scales and mathematical domains, directly enforcing value similarity via a Gaussian was inappropriate. Therefore we constructed the coupling constraint such that a higher lean towards the faster subtypes probabilistically favors higher Tau propagation rates, and vice versa. Further analysis of the coupling is detailed under Methods. We named this score a “directional potential”.

It is important to note that there was no direct empirical base to the magnitude of the score we assigned for each subtype, and the scores and weights should be tested against large-scale real patient data.

Gaussian Link 1 - Disease Stage

- **Coupled variables from ODE model:** clinical stage
- **Coupled Variable from SuStain model:** (Tau spread) expected stage, Subtypes probabilities (probability of subtype 0, probability of subtype 1, probability of subtype 2)

An increasing SuStaIn stage was strongly associated with worse global cognition and with a more advanced AD clinical diagnosis[7]. **Therefore, a higher expected Tau stage should be linked with a higher probability for a worse clinical stage** (CN to AD). Furthermore, since different subtypes experience the same Tau spread magnitude to varying degrees in different areas, the subtype also has a relation to the decrease in cognitive decline and different cognitive skills. Since in the current surrogate models we chose to focus on the memory decline (rather than language or praxis), our coupling between the Tau stage and clinical stage had to account for this

selectivity. I.e. a subtype that is related to memory-areas (subtype 1 in our case) should be related to a faster decline in earlier stages.

Therefore we created a Gaussian link between the clinical stage predicted by the ODE model, and a transformation of the SuStaIn model's Tau stage, weighted by the subtype probabilities to give a greater score for a steeper decline for patients with the appropriate decline. The transformation was required since we cannot directly link the clinical stage, which lies in the range of $[0.0, 2.0]$, to the Tau stage which lies in the range of $[1, N]$ where N is the maximal number of stages defined by the SuStaIn model.

$$\phi_2(\Theta_{ODE}, \Theta_{SuStaIn}) = \mathcal{N}(clinicalStage_{ODE}; f_2(expectedStage_{SuStaIn}, subtypeProbs_{SuStaIn}), \sigma_2)$$

Once again, the specific mathematical transformation we used (as elaborated in Methods) is not a result of a thorough research on large-scale empirical data, but more of a reasonable trial-and-error on simulated data, and can and should be revised in the future.

Gaussian Link 2 - Clinical Subtype Prior

- **Coupled variables from ODE model:** APOE4 status, Tau self rate (v_{11}), cognitive memory score
- **Coupled Variable from SuStain model:** Subtypes probabilities (probability of subtype 0, probability of subtype 1, probability of subtype 2)

While the SuStaIn model predicts the probabilities of a patient belonging to the different subtypes purely based on Tau PET brain scans, Vogel et al. showed each subtype has its own prominent characteristics, in both disease progression and patient demographics, such as the gene they carry[7]. Therefore we concluded there should

be a link between other patient characteristics given by the ODE model to the probabilities given by the SuStaIn model.

Based on real-world findings, we defined a formula to calculate the probabilities for each subtype based on data from the ODE model - where for example carrying the APOE4 gene increases the probability for certain subtypes, and better memory abilities increase the probability of another (detailed in Methods). Then we defined a Gaussian coupling between these raw calculated probabilities and the model-predicted subtype probabilities. Thus an alignment with certain subtype characteristics increases the score where the probability given to it is greater:

$$\phi_3(\Theta_{ODE}, \Theta_{SuStaIn}) = \mathcal{N}(\text{subtypeProbs}_{SuStaIn}; f_3(\Theta_{ODE}), \sigma_3)$$

Here as well, the formula we used to calculate the probabilities, though based on empirical findings, should be revised and modified upon testing the metamodel with large-scale data.

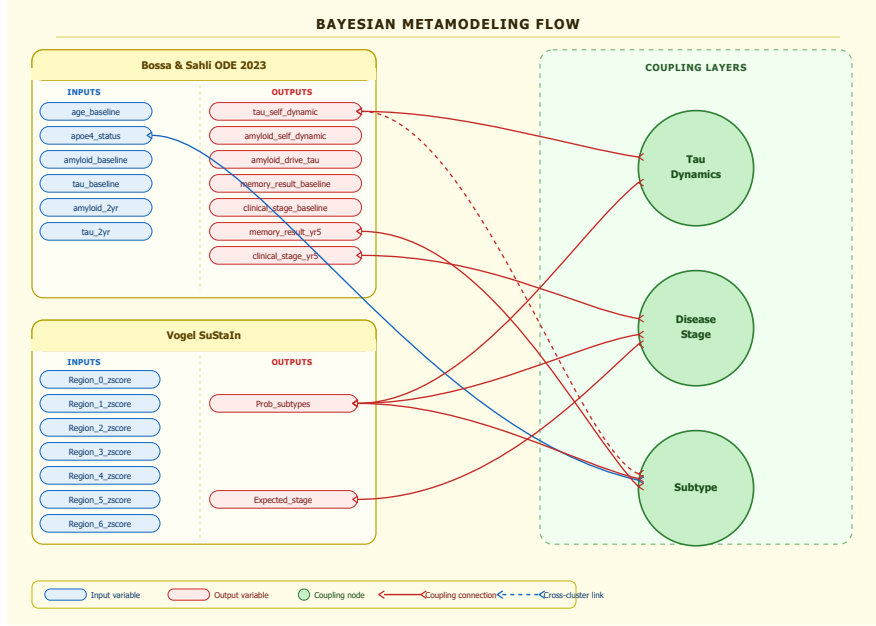


Figure 7: **Architecture of the joint Bayesian metamodel coupling the AD-ODE temporal model and the SuStaIn spatial progression model.**

(Left Box - Bossa & Sahli ODE 2023): Inputs (blue pills) comprise baseline demographics (age_baseline, apoe4_status) and 2-year longitudinal biomarker assays (amyloid, tau). Outputs (red pills) encompass kinetic rate constants (e.g., tau_self_dynamic) and 5-year cognitive/clinical trajectories (memory_result_yr5, clinical_stage_yr5).

(Left Box - Vogel SuStaIn): Inputs (blue pills) consist of cross-sectional regional Tau PET Z-scores across 7 brain regions (Region_0 to Region_6). Outputs (red pills) include spatial subtype probabilities (Prob_subtypes) and overall progression severity (Expected_stage).

(Right Box - Coupling Layers): Three central probabilistic coupling nodes (green circles) integrate the two subsystems:

Tau Dynamics Node: Modifies the ODE’s continuous tau accumulation velocity (tau_self_dynamic) based on SuStaIn’s spatial subtype classification (Prob_subtypes).

Disease Stage Node: Connects SuStaIn spatial severity (Expected_stage) and subtype profile to ODE clinical diagnostic scores (clinical_stage_baseline/yr5) and memory impairment (memory_result_yr5).

Subtype Re-weighting Node: Feeds genetic risk (apoe4_status) and ODE kinetic rates back into SuStaIn’s subtype classification node (Prob_subtypes).

3. Sampling and Inferring from the Joint Coupled Framework

After we established the coupling between the two models, we used the combined metamodel to sample and infer about patients, instead of using each model separately. This was done using a Markov Chain Monte Carlo (MCMC) algorithm, specifically Random Walk Metropolis, which compiles both models and the couplings into a unified probabilistic graph and samples over it. This is because our surrogate models, although expressive, use neural networks which function as black boxes, precluding closed-form analytical tractability of the underlying distributions from us to use to our advantage.

Given patients' data $\mathbf{X}$ we want to infer about - whether we want to get either the variables, i.e. the original models outputs, with the highest probability, or to get the means, we need to sample from the joint distribution. At every step, candidate values for all model variables across both subsystems are proposed. Instead of calculating the probability score itself, the compiler calculates the total joint log-probability score (converting likelihood products into sums). Since the joint probability given the coupling is given by

$$p(\Theta_{ODE}, \Theta_{SuStaIn} | C) \propto p_{ODE}(\Theta_{ODE}) \cdot p_{SuStaIn}(\Theta_{SuStaIn}) \cdot \prod_{i=1}^3 \phi_i(\Theta_{ODE}, \Theta_{SuStaIn})$$

we get that

$$\begin{aligned} \log p(\Theta_{ODE}, \Theta_{SuStaIn}) &= \\ &= \log p_{ODE}(\Theta_{ODE}) + \log p_{SuStaIn}(\Theta_{SuStaIn}) + \sum_{i=1}^3 \log \phi_i(\Theta_{ODE}, \Theta_{SuStaIn}) + Constant \end{aligned}$$

Therefore, all there is left to do in the inference routine is to sum the individual surrogate log-priors with the coupling log-likelihood penalties/rewards. This way, the MCMC compiler balances the priors with the couplings we defined. If the surrogates predict plausible values but the coupling penalty is high, the overall score drops and this state will be rejected, driving sampling away from that parameter region and

towards regions that are both plausible and reconcile with the biological coupling we forced.

Furthermore, because MCMC samples over this unified joint distribution, our meta-model can naturally handle incomplete patient profiles. If inputs for one of the models are missing (e.g., having regional PET scans but no longitudinal clinical history), information flows bidirectionally across the couplings to infer the unmeasured model’s parameters and cross-predict its outputs.

We simulated a prediction for two patients, and we defined their data profiles to represent two distinct patient profiles: one designed to produce conflicting predictions between submodels, and another to evaluate reinforcing agreement (amplification).

Defining the patients and the Expectations

For Subject 1, we defined a direct contradiction between the spatial presentation and the temporal/genetic characteristics: A patient reflecting rapid disease progression whose spatial features nonetheless match a subtype with a slower progression (subtype 1). We also added a contradiction between the clinical stage given by the ODE which we aimed for to be not advanced, and a relatively advanced Tau progression stage from the SuStaIn model.

Using each model separately to infer, we expected the SuStaIn model to predict a high probability for subtype 1, and for the ODE model to predict a high Tau progression rate along with an early clinical stage. However, under joint metamodel inference, we expected the contradiction to reduce the confidence in the original subtype classification in favor of faster -progressive subtypes, and that simultaneously the ODE model’s clinical forecast to be shifted toward greater severity by the advanced Tau stage.

For Subject 2, we defined a profile that aligns with both models, where spatial and temporal/genetic features mutually reinforce each other. We configured the ODE input features to represent an a severe state which is expressed in high biomarkers with a fast progression that is related to a certain Tau subtype (subtype 0); and

for the SuStaIn model to have a severe Tau stage with spatial features that also correspond to that subtype.

We expected the coupling metamodel to yield consistent directional predictions yet amplified. That is - increase the probability of severe pathological scores and the posterior certainty/probability for subtype 0, as well as lower the variance for other parameters that align between the two models, since they reinforce each other.

The results of the uncoupled models along with the **expected** results of the metamodel are presented below in Figure 8.

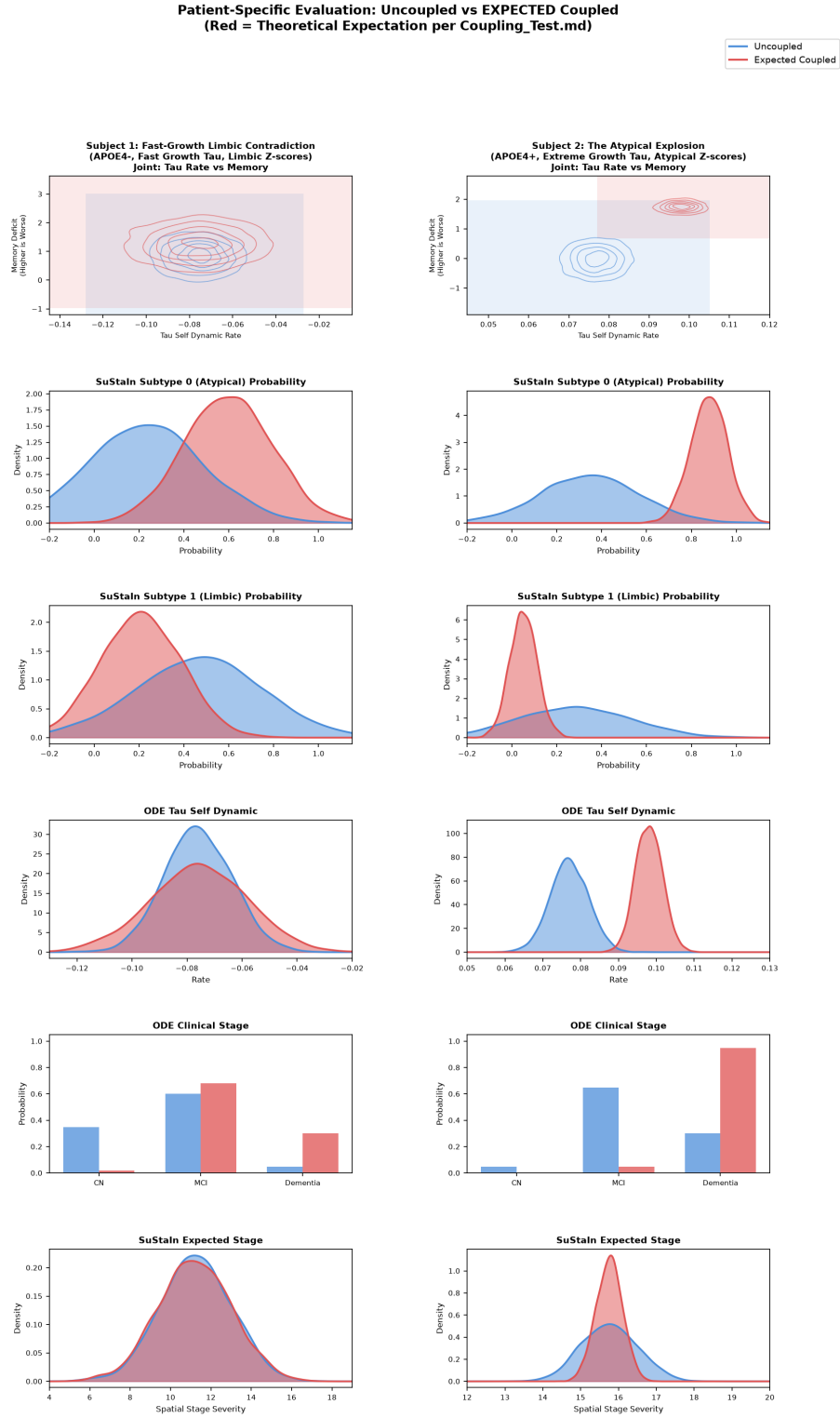


Figure 8: Theoretical expected behavior of coupled joint distributions (red) relative to uncoupled surrogate baselines (blue) across two distinct clinical test profiles.

Figure 8:

(Left Column - Subject 1: Fast-Progression Limbic Contradiction / Conflict Resolving): Patient profile with fast biological Tau growth rate, APOE4-negative status, and Limbic-predominant PET Z-scores.

Row 0: Joint 2D distribution of Tau self-dynamic rate vs. Memory deficit (higher is worse).

Row 1: SuStaIn Subtype 0 (Atypical) probability density. Expected coupling (red) surges.

Row 2: SuStaIn Subtype 1 (Limbic) probability density. Expected coupling (red) drops sharply (penalized by fast rate and APOE4- status).

Row 3: ODE Tau self-dynamic velocity (v_{11}).

Row 4: ODE Clinical Stage diagnostic probabilities (CN, MCI, Dementia). Expected coupling (red) worsens diagnostic probabilities, correcting the uncoupled naive "healthy" forecast.

Row 5: SuStaIn expected spatial progression stage.

(Right Column - Subject 2: Mutual Amplification):

Patient profile with maxed-out neocortical PET Z-scores, extreme Tau velocity, and APOE4-positive status.

Row 0: Joint 2D distribution showing variance collapse into a sharp point cluster.

Row 1: SuStaIn Subtype 0 (Atypical) probability density. Expected coupling (red) surges with a sharp spike.

Row 2: SuStaIn Subtype 1 (Limbic) probability density. Expected coupling (red) collapses to with a sharp spike.

Row 3: ODE Tau self-dynamic velocity (v_{11}). Expected coupling (red) amplifies further right with sharp variance collapse.

Row 4: ODE Clinical Stage diagnostic probabilities. Expected coupling (red) escalates to almost entirely Dementia.

Row 5: SuStaIn expected spatial progression stage.

After testing the subjects with **our metamodel**, some expectations were met and some weren't, as is described in the following figure 9.

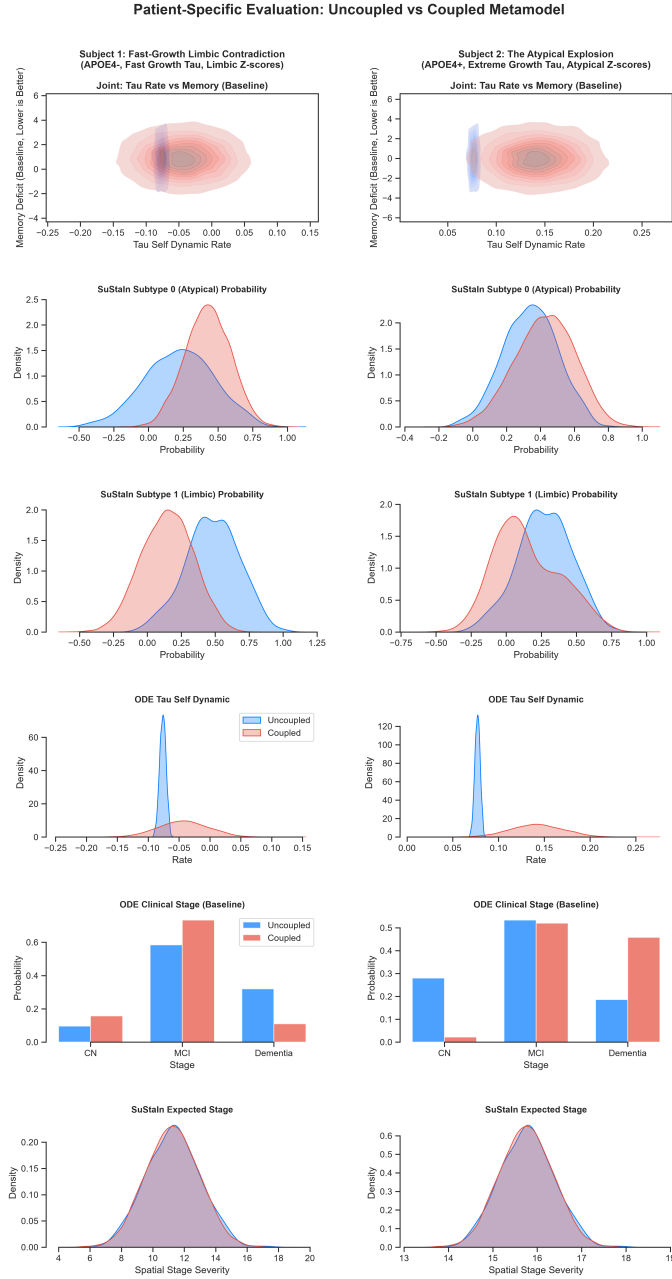


Figure 9: **Empirical MCMC coupling evaluation comparing uncoupled surrogate baselines (blue) to actual coupled metamodel posterior distributions (red).**

Figure 9: (Left Column - Subject 1: Fast-Progression Limbic Contradiction / Conflict Resolving): Patient profile with fast biological Tau growth, APOE4-negative status, and Limbic PET Z-scores.

Row 0: Joint 2D distribution of Tau self-dynamic rate vs. Memory deficit.

Row 1: SuStaIn Subtype 0 (Atypical) probability density (surged from 22.3%→42.6%).

Row 2: SuStaIn Subtype 1 (Limbic) probability density (dropped from 47.8%→14.6%, flattened peak).

Row 3: ODE Tau self-dynamic velocity (v_{11}) (widened and shifted -0.076→-0.043)

Row 4: ODE Clinical Stage diagnostic probabilities (Shifted heavily towards MCI - 73% while Dementia dropping 11%) .

Row 5: SuStaIn expected spatial progression stage (almost unchanged ~ 11.5).

(Right Column - Subject 2: Mutual Amplification): Patient profile with maxed-out neocortical PET Z-scores, extreme Tau velocity, and APOE4-positive status.

Row 0: Joint 2D distribution showing kinetic acceleration.

Row 1: SuStaIn Subtype 0 (Atypical) probability density (increased from 34.0%→42.5%).

Row 2: SuStaIn Subtype 1 (Limbic) probability density (successfully collapsed from 27.2%→14.8%).

Row 3: ODE Tau self-dynamic velocity (v_{11}) (successfully amplified right +0.077→+0.142).

Row 4: ODE Clinical Stage diagnostic probabilities (escalated to Dementia dominance 18%→46% and CN collapsing from 28%→2%).

Row 5: SuStaIn expected spatial progression stage (almost unchanged ~ 15.5).

Successes:

Clinical Stage (Subtype 2): In Subject 2, the severe spatial staging and tau acceleration escalated clinical severity to Dementia dominance.

Conflict Resolution: Fast Tau growth and APOE4-negative status in Subject 1 properly flattened Limbic confidence and the mean probability, and shifted probability toward Atypical Subtype 0.

Velocity Amplification: In Subject 2, neocortical spatial saturation successfully accelerated ODE Tau progress rate.

Phenotype Stratification: In Subject 2, subtype probabilited successfully shifted towards the Atypical Subtype 0 and away from Limbic Subtype 1.

Divergences:

Clinical Stage (Subtype 1): Despite expecting the data from the SuStaIn model to push the unconstrained ODE model toward a more severe stage, it seems posterior density concentrated more toward MCI while reducing the probability to Dementia.

Sampling Dispersion: Soft coupling tolerances and Random Walk Metropolis sampling introduced posterior dispersion rather than variance collapse - the confidence did not increase in regions of anticipated agreement.

Discussion

In this work, we aimed to overcome the limitations of domain-specific modeling of Alzheimer’s Disease by building a unifying, adaptive Bayesian metamodel capable of fusing different dimensions of the disease. As our proof of concept, we sought to couple a temporal kinetic model (the ODE model) with a subtype and staging classifier model (the SuStaIn model) to generate more robust and individualized clinical forecasts.

We successfully established a framework for this Bayesian metamodel by first converting both distinct models into fast probabilistic emulators using Simulation-Based Inference and Neural Posterior Estimators. These emulators replicated the original models with high - though not complete - fidelity at a fraction of their computational cost, critical for evaluating all sub-models simultaneously. Next, we created a joint probabilistic space using probabilistic coupling links to enforce bidirectional information flow, allowing the sub-models to influence one another toward a more holistic assessment. While tests on simulated data mostly yielded more satisfactory predictions, some deviations also highlighted important model behaviors. Overall, our metamodel proved dynamic, modular and adaptive; and given the straightforward mathematics for modifying sub-models or adding constraints, this framework can be easily expanded, successfully verifying our proof of concept.

Simulated tests demonstrated our framework’s behavior under different conditions. With Subject 1, the metamodel successfully integrated conflicting data across sub-models, shifting predictions to better satisfy the overall data profile. Subject 2 demonstrated that the couplings can also reinforce existing predictions when the sub-models align. However, some coupling effects were unexpected. For Subject 1, the contradictory data failed to alter the clinical stage towards a more severe state. Also, prediction confidence decreased not only for the conflicting Subject 1 (as expected), but also for the aligned Subject 2. This suggests that the metamodel’s use of soft coupling tolerances and the joint sampling procedure inherently introduce dispersion into the posterior distributions.

These findings indicate important limitations of our framework. While coupling is straightforward when sub-models share the same empirical variable, coupling different variables, as done here, is non-trivial. The models’ integration benefits joint inference only when guided by established domain knowledge. Furthermore, even established coupling links require careful tuning, which cannot always be determined a priori. This tuning applies both to the mathematical transformations linking the variables across sub-models and to the allowed magnitude of disagreement between models in each probabilistic link. Therefore, coupling constraints must be rigorously tested and calibrated on empirical training data before being applied broadly for future predictions.

Since this proof of concept utilized simulated data, future work should recalibrate our coupling transformations using empirical patient data. This requires comprehensive data for the same patients across all unified sub-models. Future research can also expand this framework by adding sub-models that capture AD from more dimensions, such as brain connectivity or individual cellular processes. To facilitate this expansion, there is a need to deepen our understanding of the underlying biological and mathematical connections between the diverse AD domains, alongside utilizing additional sub-models that share straightforward or overlapping connections. This will allow the framework to rely more heavily on simpler couplings, thereby minimizing complex hyperparameter tuning and reducing uncertainty.

Finally, another future direction lies in utilizing the metamodel as an exploratory research tool. By observing the resulting joint distribution on known data, researchers could explore hidden relationships across different dimensions of the disease, and even infer new coupling links beneficial for future predictions. Additionally, because of the framework’s probabilistic and bidirectional Bayesian nature, it can act as a cross-domain simulator, by taking an established phenotype from one dimension and using the metamodel to predict how it is expected to behave in other domains, while subsequently comparing it to empirical results. Ultimately, this can help identify richer, multi-dimensional AD phenotypes, hopefully beneficial for the development of better treatments for the disease.

Materials And Methods

In this section, we present the technical methodology and computational framework used to construct, couple, and evaluate the Bayesian metamodel. All models were implemented in Python. All source code, simulation scripts, surrogate training configurations, compiled coupling specifications, and evaluations are publicly available on the project’s GitHub repository at https://github.com/amitm/AD_Metamodeling.

Below, we detail (1) the sub-models used, (2) the coupling formulation, and (3) the joint sampling procedure.

1. The Sub-models

Bossa and Sahli ODE Disease Progression Model[6]

The original ODE model uses longitudinal patient data to map the continuous-time cascade of Alzheimer’s disease biomarkers via differential equations.

Model Variables, Inputs and Outputs:

The inputs to the ODE model consist of baseline patient features - specifically patient Age, binary APOE4 genetic status, and initial baseline biomarker/cognitive measurements:

$$x(0) = [A\beta, \textit{Tau}, \textit{Language}, \textit{Memory}, \textit{Praxis}]$$

The continuous temporal progression of these five latent disease burden variables is governed by a linear system of ordinary differential equations:

$$\frac{dx(t)}{dt} = A \cdot (x(t) - v_0)$$

where v_0 is an offset vector and A is the 5×5 patient-specific rate velocity matrix computed as

$$A = v + \textit{Age} \cdot w_{\textit{Age}} + \textit{APOE4} \cdot w_{\textit{APOE}}$$

v represents the baseline 5×5 interaction matrix (where diagonal elements v_{ii} dictate intrinsic self-accumulation rates and off-diagonal elements v_{ij} dictate driving forces of i on j , such as Amyloid driving Tau accumulation), while w_{Age} and w_{APOE} are demographic and genetic sensitivity matrices that modulate these accumulation rates based on patient age and APOE4 status.

By solving this linear system analytically using matrix exponentials, the model generates three primary types of outputs:

- (1) continuous multi-year trajectory forecasts for future biomarker levels and cognitive component scores (such as 2-year and 5-year Memory deficits),
- (2) inferred personalized kinetic rate constants extracted directly from matrix v (such as tau_self_dynamic v_{11} , and
- (3) diagnostic category probabilities for Cognitively Normal, MCI, and Dementia, summarized into a continuous expected clinical stage score $CS \in [0.0, 2.0]$ using an ordered logistic GLM.

How The Model Works:

Training (Learning the Equations): During the initial training phase (via MCMC), the model infers a posterior probability distribution for every coefficient in the differential equations. This is done on a database of AD patients as large as possible, where each having at least two longitudinal assessments separated by intervals of months or years. Each coefficient quantifies the influence of one biological variable on another.

We then solved the ODE system forward in time via matrix exponentials to simulate 2-year and 5-year biomarker trajectories. These synthetically generated trajectories established our baseline model dynamics and were later used as the dataset for surrogate model training.

Causal Constraints: The researchers explicitly built biological facts into these equations. For example, the model dictates that Amyloid Beta drives Tau pathology, and Tau pathology drives Memory decline. Importantly, it enforces that the reverse is not true (e.g., worsening memory does not cause Tau to accumulate; the coefficient for that is constrained to zero).

Personalization: Once the population-level coefficient distributions are established, the model can take a specific patient’s current data and fit an updated, personalized parameter distribution just for them.

Prediction: Only after calculating these personalized differential equation parameters can the model simulate the patient’s trajectory forward in time to output forecasts for clinical classification and future biomarkers.

Note: While Bossa and Sahli originally trained their population differential equation parameters using real longitudinal patient data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort, our project relied on synthetically generated patient profiles. Rather than using real clinical patient records, we synthetically constructed 4,000 patient profiles by randomly sampling baseline features (uniform Age between 60–90 and binary APOE4 status) and drawing initial baseline states (c_0) from pre-computed population posterior parameter matrices $(v, w_{Age}, w_{APOE}, v_0)$.

To compute the clinical stage CS , the model uses a linear predictor η which represents the patient’s overall disease load, calculated as a weighted combination of their biomarker values:

$$\eta = \sum_{j=1}^5 x_j(t) \cdot \beta_j$$

Using this linear predictor, the model calculates the probability of the patient belonging to each of three ordered clinical categories (1 = Cognitively Normal (CN), 2 = Mild Cognitive Impairment (MCI), 3 = Dementia (AD)) using cutpoint thresholds c_1 and c_2 :

$$P(CN) = \text{sigmoid}(c_1 - \eta)$$

$$P(MCI) = \text{sigmoid}(c_2 - \eta) - \text{sigmoid}(c_1 - \eta)$$

$$P(AD) = 1 - \text{sigmoid}(c_2 - \eta)$$

Then, to represent this probabilistic classification as a continuous variable suitable

for normalizing flows and regression links, the model computes the Expected Clinical Stage Score:

$$CS = 0 \cdot P(CN) + 1 \cdot P(MCI) + 2 \cdot P(AD)$$

This score smoothly maps the patient’s condition to the clinical categories:

- $[0.0, 0.5]$ CN
- $(0.5, 1.5]$ MCI
- $(1.5, 2]$ AD

Vogel et al. Subtype and Staging Inference (SuStaIn) Model[7]

The original SuStaIn model uses longitudinal or cross-sectional patient data to map the temporal progression of disease biomarkers into distinct trajectories (subtypes). It models how different regions of the brain accumulate pathology (e.g., Tau) over time.

Model Variables, Inputs and Outputs:

The SuStaIn model, in general, regardless to the specific AD use, is a spatiotemporal disease progression framework designed to untangle disease heterogeneity from cross-sectional data. Its inputs consist of continuous regional biomarker measurements—here specifically Tau PET Z-scores $z = [z_1, \dots, z_N]$ measured across N distinct brain regions (such as the Entorhinal cortex, Precuneus, and Amygdala). Disease progression within each subtype $s \in \{0, \dots, S - 1\}$ is modeled as an ordered sequence of discrete z-score threshold-crossing events (e.g., Z-scores reaching 1.0, 2.0, or 3.0 in specific regions), dividing spatial advancement into N_{stages} sequential stages, where N_{stages} is a hyperparameter.

Given a patient’s Z-score scan $\mathbf{z}$ SuStaIn evaluates the likelihood $P(\mathbf{z}|s, k)$ of observing that spatial pattern at stage k of subtype s . Applying Bayes’ rule with learned population subtype prevalence fractions f_s and a uniform prior over stages, the model

computes a 2D Joint Posterior Probability Matrix:

$$P(s, k | \mathbf{z}) = \frac{P(\mathbf{z} | s, k) \cdot f_s}{\sum_{s'=0}^{S-1} \sum_{k'=0}^{N_{stages}} P(\mathbf{z} | s', k') f_{s'}}$$

Summing this matrix across stages yields the Subtype Probabilities, while summing across subtypes yields the Stage Probabilities. To enable compatibility with flow-based surrogate neural networks, we collapsed the discrete stage vector into a single continuous scalar Expected Stage:

$$ExpectedStage = \sum_{k=0}^{N_{stages}} k \cdot P(Stage \ k)$$

How The Model Works:

Model training is performed on un-staged cross-sectional population datasets using a hierarchical optimization framework that combines Expectation-Maximization (EM) clustering with Markov Chain Monte Carlo (MCMC) sampling. The EM step finds maximum-likelihood estimates for the regional event sequence of each subtype and their population prevalence fractions f_s . Subsequently, MCMC sampling draws thousands of candidate event sequences and fractions to quantify model uncertainty, while Cross-Validation Information Criterion (CVIC) selects the optimal number of subtypes S . During inference on a new patient scan $\mathbf{z}$, SuStaIn bypasses trajectory optimization and directly evaluates the conditional likelihood $P(\mathbf{z} | s, k)$ against the pre-trained MCMC sequence samples, returning instant, patient-specific subtype classification probabilities and spatial stage distributions.

Note: For our project implementation, rather than using restricted empirical PET scan databases, we utilized the official synthetic data simulator provided in the authors' pySuStaIn open-source repository[7]. The simulator generates cross-sectional Z-score measurements across seven key brain regions: the Entorhinal cortex, Amygdala, Parahippocampal gyrus, Fusiform gyrus, Inferior Temporal gyrus, Middle Temporal gyrus, and Precuneus. Because each of the seven regions is evaluated across three distinct Z-score thresholds (1.0, 2.0, and 3.0), the progression timeline is discretized

into exactly 22 total spatial stages (0 to 21), where Stage 0 represents a healthy baseline and Stage 21 represents complete spatial saturation across all regions.

We fitted the pySuStaIn algorithm across candidate models ranging from 1 to 3 subtypes and evaluated model performance using Cross-Validation Information Criterion (CVIC). The 3-subtype model achieved the lowest CVIC score, proving mathematically optimal for capturing the spatial variance of the dataset. The algorithm identified three distinct tau progression trajectories:

- **Subtype 0 - Neocortical-Leaning/Atypical** (36.3% population prevalence): An atypical pathway originating outside the limbic center in the Precuneus and temporal neocortex, clinically linked to early executive and visuospatial deficits.
- **Subtype 1 - Classic Limbic** (49.5% population prevalence): The traditional Braak pathway originating in the medial temporal lobe (Entorhinal cortex and Amygdala), characterized by early memory impairment and a slower, plateauing overall progression rate.
- **Subtype 2 - Cortical-Leaning Variant** (14.2% population prevalence): A secondary neocortical variant originating in the Middle Temporal gyrus that spreads rapidly across the cortex, exhibiting fast accumulation kinetics similar to Subtype 0.

Implementing the Surrogate Models

Before surrogate training begins, the computational simulator (whether the ODE differential equation solver or the pySuStaIn algorithm) is run over thousands of forward simulations across representative physiological parameter ranges. This creates a tabular training dataset consisting of paired input-output vectors:

$$\mathcal{D} = \left\{ \left(\mathbf{x}^{(1)}, \boldsymbol{\Theta}^{(1)} \right), \dots, \left(\mathbf{x}^{(N)}, \boldsymbol{\Theta}^{(N)} \right) \right\}$$

where

- **$\mathbf{x}$ conditioning inputs:** Independent variables, initial conditions, baseline measurements, or demographic drivers (e.g., patient age, APOE4 status, or baseline regional Z-scores).
- **Θ target outputs:** Dependent outcomes, rate parameters, future forecasts, or classification probabilities (e.g., kinetic rate constants, 5-year memory deficit, spatial subtype probabilities, or expected stage).

Surrogate training is executed using the `bayesian_metamodeling` (`bayemm`) library framework written by Dr. Barak Raveh[10], via its `sbi_npe` (Simulation-Based Inference via Neural Posterior Estimation) backend.

Instead of fitting a standard deterministic regression model (which would predict only a single average value and lose uncertainty), the framework fits a deep generative neural network that models the entire conditional probability density $p(\Theta|\mathbf{x})$.

NPE is a class of Simulation-Based Inference (SBI) algorithms designed to estimate conditional probability distributions directly from simulated dataset pairs without requiring an explicit, tractable analytical likelihood function from the simulator.

To represent complex, multi-modal, non-Gaussian probability distributions $p(y|x)$, NPE utilizes Normalizing Flows, and the specific architecture we used with our backend is a Masked Autoregressive Flow (MAF). It factorizes the joint conditional density of the outputs into a product of 1D conditional distributions using the chain rule of probability. To compute this efficiently in a single forward neural network pass, MAF applies binary masks to the weight matrices of standard feedforward neural networks (Masked Autoencoder for Distribution Estimation, MADE).

The neural network weights θ are trained by minimizing the negative log-likelihood (NLL) over the dataset using Adam optimization:

$$\mathcal{L}(\Theta) = -\frac{1}{N} \sum_{i=1}^N \log p_{MAF}(\Theta^{(i)}|\mathbf{x}^{(i)}; \theta)$$

Once NLL loss converges, the backend library `bayesmm` serializes the trained neural

network weights, feature normalization scalers, and input/output schemas into json files. During joint metamodel sampling, these saved artifacts are loaded as drop-in probabilistic emulators, enabling instant density evaluation and sample generation.

ODE Model Surrogate Training

To train the ODE surrogate model we constructed a 4,000-sample dataset by integrating the differential equations forward in time using matrix exponentials from pre-computed MCMC parameter posteriors (found in `ad_ode_demo_posterior.pkl`). The inputs and outputs we selected for the surrogate were:

- **inputs:** `age_baseline` (sampled uniformly between 60–90 years), binary `APOE4_status` (Bernouli 0.5), and 2-year longitudinal biomarker measurements (`amyloid_baseline`, `tau_baseline`, `amyloid_2yr`, `tau_2yr`)
- **outputs:** 2 of the intrinsic ODE kinetic rate parameters (`tau_self_dynamic  $v_{11}$` , `amyloid_self_dynamic  $v_{00}$` , `amyloid_drive_tau  $v_{10}$`), `baseline_memory_score`, 5-year forecasted memory score (`memory_result_5yr`), and clinical stage scores (`clinical_stage_baseline` and `clinical_stage_5yr`).

SuStaIn Model Surrogate Training

To train the SuStaIn surrogate model, we constructed a 500-sample dataset generated by running pySuStaIn’s MCMC classification engine over synthetic regional scan profiles (found in `tau_sim_10k_simulated_data.pickle`). The inputs and outputs we selected for the surrogate were:

- **Inputs:** Continuous Tau PET Z-scores across 7 brain regions (`region_0_zscore` through `region_6_zscore`).
- **Outputs:** Marginal posterior probabilities for belonging to each discovered spatial trajectory (`prob_subtype_0`, `prob_subtype_1`, `prob_subtype_2`), and the continuous spatial disease stage (`expected_stage`).

2. The Coupling

We bridged the two models across three core domains using custom mathematical transformations as detailed below. Note that the metamodeling framework requires to define each coupling as having a source with some variables and a target with other variables, but the bidirectional property of the probability function and samplings makes this a matter of pure notation which helps to distinguish the two sides we couple.

1. Subtype-Drive Velocity/Tau Dynamics

- **Source (SuStaIn):** Probabilities of the different subtypes (`prob_subtype_0`, `prob_subtype_1`, `prob_subtype_2`)
- **Target (ODE):** Tau self dynamic rate v_{11} (`tau_self_dynamic`)

We defined a `velocity_modifier_score` transform that generates a modifier score S using a weighted (W) sum of the different subtype probabilities (P)

$$S = \sum_{i=0}^2 (P_i \times W_i)$$

Where we set the weights such that the sum is equal to

$$S = (P_0 \cdot 1.8) + (P_1 \cdot (-0.6)) + (P_2 \cdot 1.0)$$

Notice that the weights for subtypes W_0, W_2 are positive to indicate their relation to a different higher rate of Tau spread, whereas the weight for subtype 1 (the classic lymbic) W_1 is negative to indicate a relation to a slower rate.

We wanted the link to have a linear directional potential, such that higher probabilities for subtypes 0 and 2 favor higher Tau rates, and a higher probability for subtype 1 will favor a lower rate. Since the sampling and inference algorithms on the joint

distribution use log-likelihoods and logarithms of the probabilities, we chose the link coupling constraint formula to be

$$\begin{aligned} \log p(\text{Tau_Dynamic_Coupling} | \Theta_{ODE}, \Theta_{SuStaIn}) &= \log \phi_1(\Theta_{ODE}, \Theta_{SuStaIn}) \\ &= \frac{1}{\sigma_1} \times (\text{tau_self_dynamic} \times S) \end{aligned}$$

Where $\sigma_1 = 0.6$ is the magnitude modifier - a larger σ_1 corresponds to a weaker reward/punishment.

Since the log score is added to the other log probabilities, when comparing between two tau_self_dynamic it is clear that a higher rate with a negative modifier score S punishes the model whereas a positive score S rewards it, and vice versa for a slower rate.

2. Disease Stage

- **Source (SuStaIn):** The calculated Tau expected_stage weighted by the subtype probabilities (prob_subtype_0, prob_subtype_1, prob_subtype_2)
- **Target (ODE):** The continuous single clinical_stage

In order to translate the SuStaIn's Tau expected stage, which is on a discrete range of 0-21 to the continuous variable in the ODE that tracks the clinical stage continuously and is range of [0.0, 2.0] we had to define a transformation. Since the stage is simply a measure of how progressive the Tau spread in the patient's brain and not a direct measure of the clinical stage, different subtypes can have different clinical stages that are derived from the same expected stage. So our transformation took the subtype probabilities into consideration.

Then we linked the clinical stage of the ODE to the transformed Tau expected stage using a Gaussian connection to reward outputs that sustain the relation we defined. The transformation and the Gaussian link are detailed below.

Let x be the SuStaIn's Tau expected_stage. We defined two logistic progression curves targeting a max clinical severity of 2.0. One for Subtype 1 (limbic) and one for subtypes 0,2 which had more similar characteristics in this regard.

- **Progression curve for Subtype 1:** rises faster at lower stages (midpoint anchor 9.5), since this subtype is related to a faster clinic decline in earlier Tau stages.

$$C_{limbic}(x) = \frac{2.0}{1 + e^{-\alpha(x-9.5)}}$$

- **Progression curve for Subtypes 0, 2:** rises slower, reaching dementia at higher stages (midpoint anchor 13.5).

$$C_{Atypical}(x) = \frac{2.0}{1 + e^{-\alpha(x-13.5)}}$$

Where $\alpha = 0.4$ is a hyperparameter for the progression rate.

The final expected clinical stage which is the transformation result is

$$\text{Expected Clinical Stage (ECS)} = P(S_1) \cdot C_{limbic}(x) + (P(S_0) + P(S_2)) \cdot C_{Atypical}(x)$$

And thus the final coupling link is given by the Gaussian link:

$$\begin{aligned} p(\text{clinical_stage} | \Theta_{ODE}, \Theta_{SuStaIn}) &= \phi_2(\Theta_{ODE}, \Theta_{SuStaIn}) \\ &= \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(\text{ECS} - \text{expected_stage})^2}{2\sigma_2^2}\right) \end{aligned}$$

where we set $\sigma_2 = 0.15$.

Once again, since the sampling and inference use logs, we get that

$$\log \phi_2(\Theta_{ODE}, \Theta_{SuStaIn}) = -\frac{1}{2\sigma_2^2} (\text{ECS} - \text{expected_stage})^2 + \text{Constant}$$

And it is easy to see that the more the two variables are similar, the smaller the

punishment to the probability assessment.

3. Subtype Prior

- **Sources (ODE):** Disease progression parameters and results (APOE4_status, tau_self_dynamic, memory_cognitive_test_result).
- **Targets (SuStaIn):** Subtype probabilities (prob_subtype_0, prob_subtype_1, prob_subtype_2)

Here we wanted to incorporate the patient’s clinical history and characteristics as another prior to set the probabilities of the Tau trajectory subtypes. Using known empirical associations (carrying APOE4 gene and earlier severe memory deficit favors the limbic subtype, rapid decline and favors the other subtypes), we calculated scores for each subtype group: S_{limbic} for subtype 1, and $S_{Atypical}$ for subtypes 0,2. These scores were then passed through a softmax function to generate a probability distribution that acts as a constraint on the SuStaIn model’s own subtype predictions - which we linked using a Gaussian link.

First, we map the continuous rate of tau spread tau_self_dynamic v_{11} to $[0, 1)$ using a sigmoid:

$$V_{norm} = \frac{1}{1 + e^{-1.2v_{11}}}$$

Next, we clip the memory test score memory_cognitive_test_result M to ensure it is within $[0, 1]$

$$M_{norm} = clip(M, 0, 1)$$

Then, using also the binary APOE4_status A , we calculated the scores for limbic and atypical phenotypes. Notice that a higher memory score corresponds to a more

damaged memory, therefore it is added to the limbic score:

$$S_{limbic} = \alpha_1 A + \alpha_2 (1 - V_{norm}) + \alpha_3 M_{norm}$$

$$S_{Atypical} = \alpha_1 (1 - A) + \alpha_2 (V_{norm}) + \alpha_3 (1 - M_{norm})$$

where we set $\alpha_1 = 0.25$ and $\alpha_2 = \alpha_3 = 1$ are hyperparameters.

So we have raw scores $RS = [S_{Atypical}, S_{limbic}, S_{Atypical}]$

Finally, we set the softmax:

$$CoupledP(Subtype_i) = \frac{\exp(\beta \cdot RS[i])}{\sum_{j=0}^2 \exp(\beta \cdot RS[j])}$$

where $\beta = 2$ is a temperature hyperparameter.

This result is linked to the SuStaIn's original prediction for the probabilities using a Gaussian just like for the transformed Tau expected stage, so overall we get

$$\begin{aligned} p(\text{prob_subtype_i} | \Theta_{ODE}, \Theta_{SuStaIn}) &= \phi_{3_i}(\Theta_{ODE}, \Theta_{SuStaIn}) \\ &= \frac{1}{\sqrt{2\pi\sigma_3^2}} \exp\left(-\frac{(\text{CoupledP}(\text{Subtype}_i) - \text{prob_subtype_i})^2}{2\sigma_3^2}\right) \end{aligned}$$

where we set $\sigma_3 = 0.15$.

3. Joint Inference

For the joint sampling and the inference process the metamodeling framework compiles the individual surrogates (SurrogateLikelihoodFactorIR) and cross-model coupling links (CouplingFactorIR) into a single, unified probabilistic graph using PyMC. It treats both the surrogate models and the coupling links as intermediate representations that assess samples' likelihood simultaneously, and combines them to accept or reject sampling suggestions.

The metamodeling specifications (surrogate models, variables, coupling links) are saved in a designated high-level .json file. It is then compiled into an explicit Bayesian factor graph where the observed (input) variables are clamped to their values, and the latent (output) parameters form the active parameter space to be sampled: $\Theta = [\theta_{ODE}, \theta_{SuStaIn}]$.

For any candidate state Θ proposed during inference, the compiled graph calculates the total joint log-density under the separate items:

$$\log \mathcal{L}(\Theta) = \log p_{ODE}(\theta_{ODE}) + \log p_{SuStaIn}(\theta_{SuStaIn}) + \sum_{i=1}^3 \log \phi_i(\theta_{ODE}, \theta_{SuStaIn})$$

Importantly, since the surrogate models currently use NPE without symbolic gradients for the sampler to use, we use a version of an Adaptive Component-Wise Random-Walk Metropolis Sampler. This allows exploring effectively the parameter space by rejecting/accepting candidate samples until the satisfied amount and regions of posterior draws are drawn.

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